INFO 6105 – Data Science Engineering Methods and Tools

Final Project Report: PhiUSIIL Phishing URL

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Introduction

In the digital age, phishing remains one of the most pervasive and dangerous forms of cyberattacks. Attackers often disguise malicious URLs to mimic legitimate websites, tricking users into revealing sensitive information such as passwords, credit card details, and other personal data. Despite significant efforts in cybersecurity awareness and user training, phishing attacks continue to evolve, making automated detection techniques more crucial than ever.

This project focuses on exploring patterns and identifying distinguishing characteristics in phishing URLs using data science techniques. By leveraging the **Phishing Website Dataset (PhiUSIIL)** from the UC Irvine Machine Learning Repository, the aim is to conduct a comprehensive **Exploratory Data Analysis (EDA)** to understand the underlying features associated with phishing websites. The dataset comprises various lexical, host-based, and content-based features derived from website URLs, each labeled as either legitimate or phishing.

The objective of this analysis is not to build a predictive model, but rather to:

- Understand the distribution and importance of different features.
- Investigate feature relationships through correlation and principal component analysis (PCA).
- Answer research questions related to the behavior and structure of phishing URLs using statistical testing and visualization techniques.

Through this approach, the project aims to generate insights that can support cybersecurity practitioners in better understanding phishing mechanisms and potentially guide future model-building efforts or real-world detection systems.

Research Questions:

1. What are the key distinguishing features between legitimate and phishing URL?
2. How do URL characteristics such as TLDLegitimateProb or URLTitleMatchScore affect the likelihood of a malicious URL?

Methodology

Data Preparation:

- Checked for missing values, data types, and class distribution to ensure data quality for accurate analysis and modeling.

- Converted the label column into a categorical factor (Legitimate vs. Phishing).
- Standardized feature names and levels for better clarity and analysis.

Features Selection:

Three key features were selected for in-depth analysis based on their potential relevance to phishing behavior:

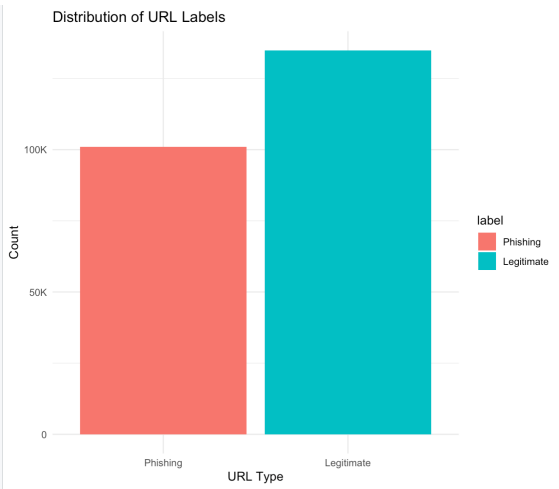
- TLDLegitimateProb: Probability that the Top-Level Domain is associated with legitimate websites.
- URLTitleMatchScore: A score indicating how closely the page title matches the URL.
- CharContinuationRate: A metric reflecting character patterns common in URLs.

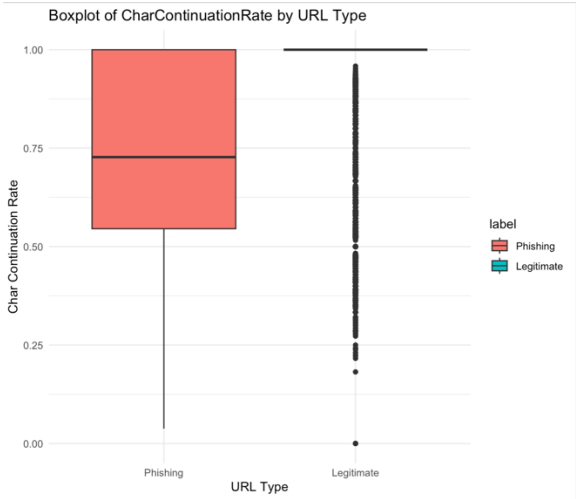
Exploratory Data Analysis (EDA):

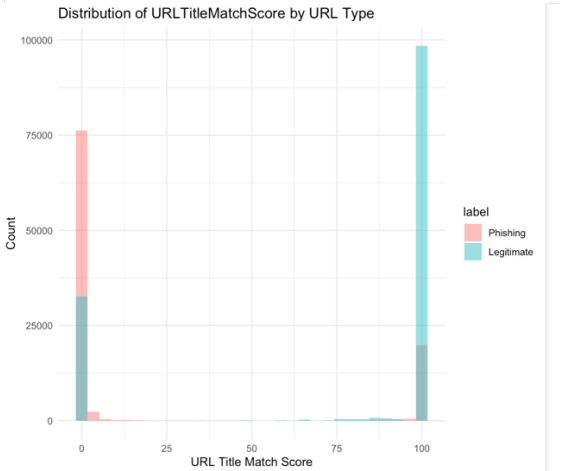
Feature	Description	Key Observation	Interpretation
Label	URL type (Phishing or Legitimate)	Legitimate: 134,850 (56.8%) Phishing: 100,945 (43.2%)	Slight class imbalance
TLDLegitimateProb	Probability that a URL's TLD (Top-Level Domain) is associated with legitimate sites	Phishing URLs are concentrated near 0.0 Legitimate URLs peak near 0.5+	Legitimate domains tend to use trusted TLDs
URLTitleMatchScore	Score indicating how well the page title matches the domain	Legitimate URLs skew toward higher scores	Phishing sites often use mismatched or deceptive titles
CharContinuationRate	Measures character pattern continuity	Legitimate URLs cluster around 1.0 Phishing URLs have more variation	Phishing URLs tend to have unnatural patterns (e.g., random characters)
Missing Values	Presence of NA in columns	None detected across dataset	Dataset is clean and ready for modeling

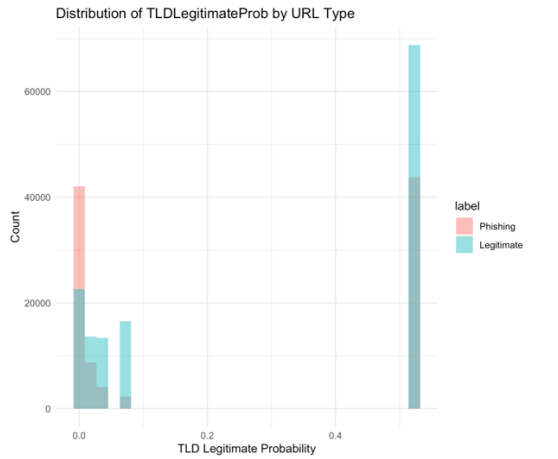
EDA was conducted to visualize and summarize the distributions of selected features across phishing and legitimate URLs. Techniques included:

- Histograms and boxplots to examine the spread and central tendency of feature values by label.
- Bar plots to visualize the class distribution of URL types.









Statistical Testing:

To determine whether the selected features significantly differ between phishing and legitimate URLs, the following tests were applied:

- Independent t-tests for each feature to compare group means.

Feature	Phishing Mean	Legitimate Mean	Difference	p-value	Statistically Significant?	Interpretation
TLDLegitimateProb	0.2321	0.2816	-0.0495	< 2.2e-16	Yes	Legitimate URLs are more likely to use trusted TLDs (like .com, .org)
URLTitleMatchScore	21.20	75.27	-54.07	< 2.2e-16	Yes	Legitimate URLs tend to have titles that closely match the domain

CharContinuationRate	0.7284	0.9332	-0.2048	< 2.2e-16	Yes	Legitimate URLs show more consistent character patterns, less randomness

- One-way ANOVA to validate significance across label groups.

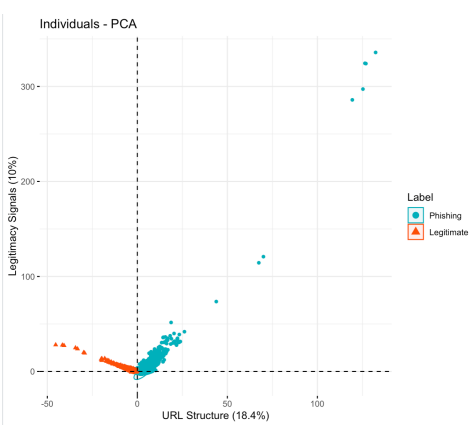
Feature	F-value	p-value	Significance
TLDLegitimateProb	2258	< 2e-16	Strong
URLTitleMatchScore	96765	< 2e-16	Extremely Strong
CharContinuationRate	66032	< 2e-16	Very Strong

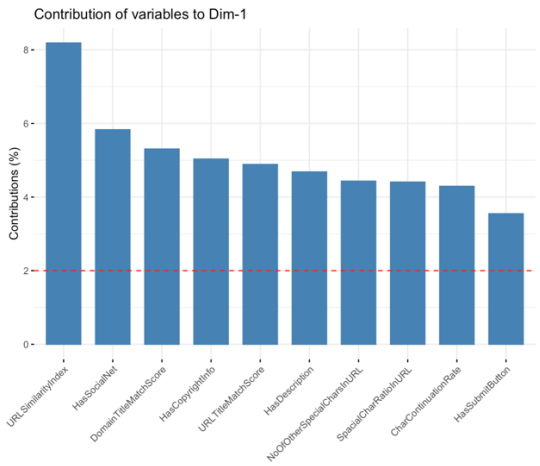
- Effect size (Cohen's d) was calculated to measure the magnitude of differences.

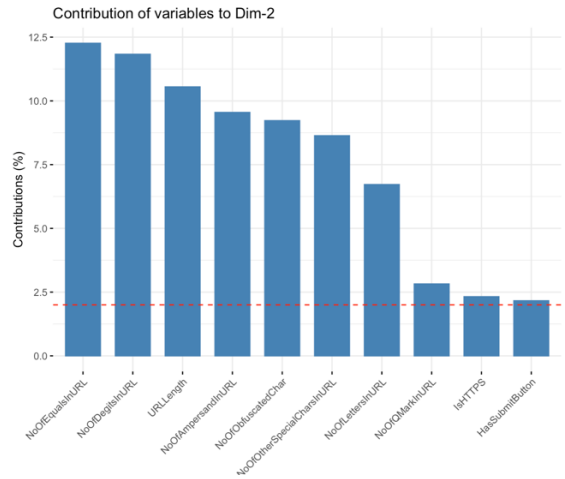
Feature	Cohen's d	Effect Size	Interpretation
TLDLegitimateProb	-0.198	Negligible	while statistically different, has a negligible effect size, indicating it may not be a strong predictor on its own.
URLTitleMatchScore	-1.295	Large	Legitimate URLs match their page titles far better than phishing URLs. This is a powerful distinguishing feature.
CharContinuationRate	-1.069	Large	Phishing URLs tend to have more randomness in character flow, while legitimate ones are more structured.

Principal Component Analysis (PCA):

- Performed PCA to reduce dimensionality and identify patterns in feature space.
- Visualized results to observe class separation and feature contributions.
- Highlighted which features contributed most to variance across URLs.





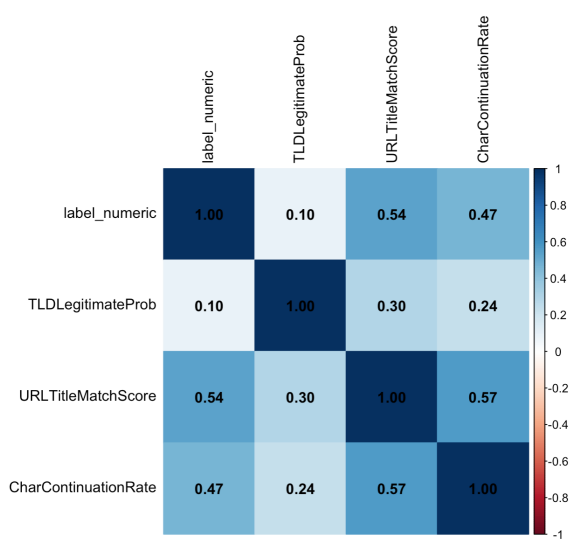


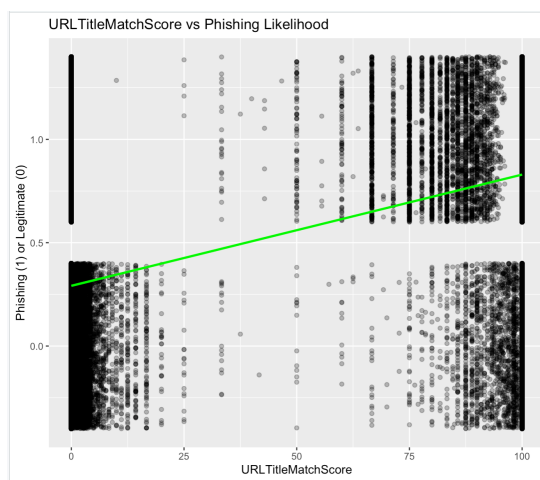
Dim1 - Legitimacy Signals (18.4%)

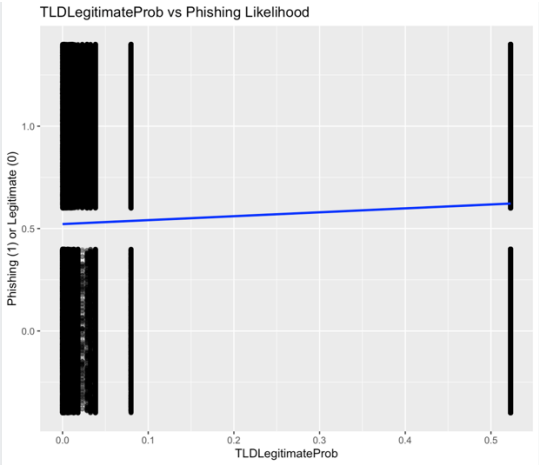
Dim2 - URL Structure (10%)

Correlation and Linear Analysis:

- Converted the label into numeric format (1 = Legitimate, 0 = Phishing) for correlation analysis.
- Generated a correlation matrix to assess relationships between selected features and phishing likelihood.
- Used scatterplots with regression lines to visualize linear trends.







Key Points

- TLDLegitimateProb values were significantly lower for phishing URLs, indicating that phishing websites often use less credible or uncommon top-level domains.

- URLTitleMatchScore was also lower for phishing URLs, suggesting a frequent mismatch between the page title and its URL—an indicator of deceptive intent.

- CharContinuationRate displayed modest differences, potentially reflecting abnormal or auto-generated character patterns in phishing links.

- Statistical tests (t-tests and ANOVA) confirmed significant differences across all three features between phishing and legitimate URLs.

- Effect size (Cohen's d) analysis highlighted TLDLegitimateProb is the most effective at differentiating between phishing and legitimate URLs

- Correlation analysis revealed a positive relationship between legitimacy and both TLDLegitimateProb and URLTitleMatchScore, meaning higher values are associated with legitimate URLs.

- Principal Component Analysis (PCA) showed effective dimensionality reduction, with clear separation between phishing and legitimate classes in reduced feature space.
  PCA results also highlighted the strong contribution of the selected features to overall variance, reinforcing their importance.

  The overall analysis highlights the effectiveness of lexical and semantic URL features in identifying phishing threats and provides a solid foundation for future machine learning or rule-based detection systems.

Conclusion

This exploratory analysis of the PhiUSIIL Phishing URL Dataset provided meaningful insights into the characteristics that distinguish phishing URLs from legitimate ones. Through statistical testing, correlation analysis, and principal component analysis, several key features—particularly TLDLegitimateProb and URLTitleMatchScore—emerged as strong indicators of phishing behavior.

The findings demonstrated that phishing URLs tend to have lower domain legitimacy probabilities and less alignment between page titles and URLs, reflecting common deception strategies. These insights were not only statistically significant but also visually and structurally supported through PCA and correlation methods.

While this project did not involve predictive modeling, the results lay a strong foundation for future work in phishing detection. The identified features can be valuable inputs for machine learning models or real-time URL filtering systems, contributing to more effective cybersecurity solutions.

References

Prasad, A., Chandra,S (2023). *PhiUSIIL Phishing URL Dataset* [Data set]. UCI Machine Learning Repository.
https://archive.ics.uci.edu/dataset/967/phiusiil+phishing+url+dataset

URL Link of Video Recording

https://drive.google.com/file/d/1sGE60HuSrFE6wBSZejpin8ERCkLMk5f9/view?usp=sharing